Inverse of HT Matrix and Chapter 2 Wrap-Up

EGR 445/545

GVSU

Plan for Today

- review
- inverse of an HT matrix
- wrap-up chapter 2
    - Euler angles
    - free vectors
    - computational considerations
- Quiz 1
- LA 6 and/or project time

What does this mean?

$$
{}^{A}_{B}\mathbf{R} = \begin{bmatrix} {}^{A}\hat{X}_B & {}^{A}\hat{Y}_B & {}^{A}\hat{Z}_B \end{bmatrix} = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}
$$

Unit Vectors and Dot Products

$${}^{B}\hat{X}_{B} = ?$$

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$$^{B}\hat{X}_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Unit Vectors and Dot Products

$$ {}^{B}\hat{X}_{B} = ? $$

$$? = {}^{A}_{B}\mathbf{R} \cdot {}^{B}\hat{X}_{B} $$

$$ {}^{B}\hat{X}_{B} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} $$

Unit Vectors and Dot Products

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Unit Vectors and Dot Products

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$$^{A}\hat{X}_{B} = {}^{A}_{B}\mathbf{R} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Unit Vectors and Dot Products (continued)

$$^{A}\hat{X}_{B} = {}^{A}_{B}\mathbf{R} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$^{A}_{B}\mathbf{R} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = ?$$

$$^{A}\hat{X}_{B} = \ ?$$

LA 5 feedback

If you are doing trig manually to find an HT matrix, you are probably doing something wrong.

- maybe not getting the wrong answer, but you are at least doing things the hard way
- you might have your origins misplaced

```
P_Tip2 = np.array([8,0,0,1])
T_12 = robotics.HTz(35,0,0,0)
T_01 = robotics.HTz(25,10*cosd(25),10*sind(25),0)
```

Finding the Inverse of an HT Matrix

Goal and Problem Statement

- assuming we know ${}^{A}_{B}\mathbf{T}$, how do we find its inverse quickly and cleanly?
- why do we need to do this?
  - what's wrong with doing it numerically?
    - i.e. `numpy.linalg.inv`
- why can't we just use the transpose like with rotation matrices?

HT Inverse Matrix (end goal)

The book throws this out without much explanation:

$$ {}^{B}_{A}\mathbf{T} = \begin{bmatrix} & {}^{A}_{B}\mathbf{R}^{T} & & -{}^{A}_{B}\mathbf{R}^{T} \cdot {}^{A}\mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{(eqn. 2.45)} $$

- I want you to understand what this means and where it comes from.

${}^{A}_{B}\mathbf{T}$ vs. ${}^{B}_{A}\mathbf{T}$

- what steps does ${}^{A}_{B}\mathbf{T}$ capture?
- if ${}^{B}_{A}\mathbf{T}$ maps the opposite direction, what steps does it represent?

Letting the notation guide us

$$ {}^{A}P = {}^{A}_{B}\mathbf{R} \cdot {}^{B}P + {}^{A}P_{BORG} $$

$$ {}^{A}_{B}\mathbf{T} = \begin{bmatrix} & {}^{A}_{B}\mathbf{R} & & {}^{A}\mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{A}P = {}^{A}_{B}\mathbf{T} \cdot {}^{B}P $$

- what does ${}^{B}_{A}\mathbf{T}$ do?
    - what equation does it represent?

Reversing the Mapping

- What does this do? What does an HT matrix combine?

$$^{B}P = {}^{B}_{A}\mathbf{T} \cdot {}^{A}P$$

$$^{B}P = {}^{B}_{A}\mathbf{R} \cdot {}^{A}P + ^{B}P_{AORG}$$

- how do we put it in matrix form?

$$\begin{bmatrix} ^{B}P \\ 1 \end{bmatrix} = \begin{bmatrix} & {}^{B}_{A}\mathbf{R} & & ^{B}\mathbf{P}_{AORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ^{A}P \\ 1 \end{bmatrix}$$

Conceptual Question

- conceptually, what is ${}^{B}P_{AORG}$?
  - where does it start and end?
  - what frame is it expressed in?
  - why is it expressed in that frame?
- what is the relationship between ${}^{A}P_{BORG}$ and ${}^{B}P_{AORG}$?

Examples of ${}^{B}P_{AORG}$

- pure translation
- translation and rotation
- 3/4/5 example

Pieces of the Reversed Mapping

- how do we find these pieces based on stuff we already know?
  - assuming we have ${}^{A}_{B}\mathbf{T}$?

$$ {}^{B}_{A}\mathbf{R} = ? $$

$$ {}^{B}\mathbf{P}_{AORG} = ? $$

Finding the Pieces of the Reversed Mapping

$$ {}^{B}_{A}\mathbf{R} = {}^{A}_{B}\mathbf{R}^{T} $$

$$ {}^{B}\mathbf{P}_{AORG} = {}^{B}_{A}\mathbf{R}(-{}^{A}\mathbf{P}_{BORG}) $$

$$ {}^{B}\mathbf{P}_{AORG} = {}^{A}_{B}\mathbf{R}^{T}(-{}^{A}\mathbf{P}_{BORG}) $$

Final Formulation

$$\begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix} = \begin{bmatrix} & {}^{A}_{B}\mathbf{R}^{T} & & -{}^{A}_{B}\mathbf{R}^{T} \cdot {}^{A}\mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix}$$

HT Inverse Matrix

$$
{}^{B}_{A}\mathbf{T} = \begin{bmatrix} & {}^{A}_{B}\mathbf{R}^{T} & & -{}^{A}_{B}\mathbf{R}^{T} \cdot {}^{A}\mathbf{P}_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{(eqn. 2.45)}
$$

which is the same as

$$
{}^{B}_{A}\mathbf{T} = \begin{bmatrix} & {}^{B}_{A}\mathbf{R} & & {}^{B}\mathbf{P}_{AORG} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix}
$$

Converting this to Python Code

```python
def HTinv(Tin):
    Tout = eye(4)
    R = Tin[0:3,0:3]
    Ri = R.T
    Tout[0:3,0:3] = Ri
    P_BorgA = Tin[0:3,3]
    P_AorgB = -1.0*dot(Ri,P_BorgA)
    Tout[0:3,3] = P_AorgB
    return Tout
```

What are Euler angles and why should we care?

- someday you may have to talk to an old, grumpy engineer
- do rotation matrices really have 9 unknowns?

$$\mathbf{R} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

- what constraints are placed on a valid rotation matrix?
  - and where do those constraints come from?

Euler Angles: What do we mean by "arbitrary rotation"?

- are rotation matrices restricted to x, y, or z axes?
- what use would we have for an arbitrary rotation matrix?

Euler Angles

- **big ideas:**
  - an *arbitrary* rotation matrix has three unknowns that can be represented using three angles
  - an *arbitrary* rotation/orientation can be expressed as the product of three rotation matrices:

$${}^{A}_{B}\mathbf{R} = \mathbf{R}(\alpha) \cdot \mathbf{R}(\beta) \cdot \mathbf{R}(\gamma)$$

- we primarily focus on rotations about a single axis
- Euler angles handle the more general case of arbitrary orientation between frames A and B

Free Vectors

What are free vectors and how to we deal with them?

- are some vectors not free?
  - is this a social justice issue?
- is there a whale or dolphin that needs help getting back to the ocean?
- how much do regular vectors cost?
  - do we get free stuff?

Free vs. Line Vectors

- a free vector can be placed anywhere
- the opposite of a free vector is a line vector
- the line of action is important for a line vector
    - you cannot move a line vector without affecting its meaning
- positions and forces are examples of line vectors
- velocities and moments are examples of free vectors

Velocities in different frames

- imagine an AGV in an Amazon warehouse
- the AGV is being observed by two different stations
  - station {B} is rotated and translated from station {A}
  - the velocity of the AGV would have different components in the two frames
  - should the magnitude of the velocity be different in the two frames?

Alternative Velocity Question

$$^{A}\mathbf{P} \ = \ ^{A}_{B}\mathbf{R}\,^{B}\mathbf{P} + ^{A}\mathbf{P}_{BORG}$$

- velocity is the derivative of position
- taking derivatives in moving reference frames is graduate dynamics
- if $^{A}\mathbf{P}_{BORG}$ is constant, what happens when we take the derivative?

Computational considerations

- what is the fastest way to compute ${}^{0}\mathbf{P}$?

$$ {}^{0}\mathbf{P} = {}^{0}_{1}\mathbf{T}\,{}^{1}_{2}\mathbf{T}\,{}^{2}_{3}\mathbf{T}\,{}^{3}\mathbf{P} $$

Computational considerations: which is faster?

- option 1: Find ${}^0_3\mathbf{T}$ first

$$
\begin{aligned}
{}^0_3\mathbf{T} &= {}^0_1\mathbf{T}\,{}^1_2\mathbf{T}\,{}^2_3\mathbf{T} \\
{}^0\mathbf{P} &= {}^0_3\mathbf{T}\,{}^3\mathbf{P}
\end{aligned}
$$

- option 2: Find ${}^2\mathbf{P}$, then find ${}^1\mathbf{P}$, and finally ${}^0\mathbf{P}$

$$
\begin{aligned}
{}^2\mathbf{P} &= {}^2_3\mathbf{T}\,{}^3\mathbf{P} \\
{}^1\mathbf{P} &= {}^1_2\mathbf{T}\,{}^2\mathbf{P} \\
{}^0\mathbf{P} &= {}^0_1\mathbf{T}\,{}^1\mathbf{P}
\end{aligned}
$$